BLOCKCHAIN-BASED PHARMACEUTICAL SUPPLY CHAIN WITH ANTI-COUNTERFEIT VERIFICATION

A PROJECT REPORT

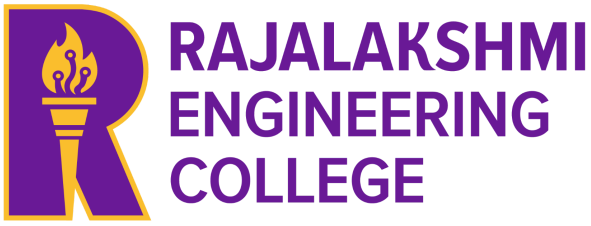
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BONAFIDE CERTIFICATE

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ABSTRACT

The Drug Inventory and Supply Chain Tracking System using Hyperledger Fabric is designed to improve transparency, traceability, and security in pharmaceutical distribution. The proposed system uses blockchain technology to maintain a tamper-resistant record of drug registration, ownership transfers, and verification throughout the supply chain. Each medicine is registered with a unique serial number, batch number, drug name, manufacturer, manufacturing date, expiry date, and quantity. Hyperledger Fabric smart contracts validate transactions and maintain consistent drug records across authorized participants in the network. A web-based interface enables users to register medicines, view inventory details, transfer ownership, verify drug authenticity, and inspect transaction history. QR codes containing unique drug serial numbers support quick identification and verification through a connected application. A Node.js and Express backend communicates with the Hyperledger Fabric network, while the frontend provides an accessible interface for managing pharmaceutical records. By recording supply chain activities on a permissioned blockchain, the system helps identify unauthorized modifications, improves accountability, and supports the detection of counterfeit or unverified medicines. The proposed framework provides a secure and transparent approach to pharmaceutical inventory management and supply chain tracking.

Keywords: Hyperledger Fabric, Blockchain, Drug Traceability, Pharmaceutical Supply Chain, Smart Contracts, Drug Verification, QR Code, Inventory Management, Transaction History, Counterfeit Drug Detection.

The system also maintains a chronological history of drug-related transactions, allowing authorized users to examine ownership changes and track medicine movement through the supply chain. Smart contracts enforce the defined rules for registering drugs, retrieving records, verifying authenticity, and transferring ownership. The integration of a web application and an Android-based verification interface improves accessibility and simplifies drug identification. By combining permissioned blockchain infrastructure, unique drug identifiers, and QR-based verification, the proposed system aims to strengthen supply chain transparency, reduce the risk of unauthorized record manipulation, and support more reliable pharmaceutical distribution

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CHAPTER 1
INTRODUCTION

1.1 MOTIVATION

India's pharmaceutical supply chain faces challenges similar to agriculture's disease problems: complex, distributed systems where counterfeit drugs, mislabeling, and poor visibility endanger patients and revenue. Medicines may pass through multiple manufacturers, distributors, and pharmacies across varied regions, and without robust tracking, falsified or expired products can slip into circulation. Existing supply-chain solutions often "lack the transparency, adaptability, and traceability" needed to stop this. In short, stakeholders need an intelligent, scalable, and privacy-conscious way to monitor every drug from creation to dispensing.

To meet this need, we built a permissioned blockchain system (Hyperledger Fabric) that registers and traces each drug unit on-chain. Every package carries a unique QR code encoding its identifiers (GTIN, serial number, batch, expiry). Scanning that QR invokes our network rather than a central database: the ledger holds the "yes/no" verification of that serial. In practice, a smartphone or scanner reads the QR and the system runs a chaincode query like readDrug or verifyDrug. Under the hood, peers from each organization (manufacturer, regulator, distributor, pharmacy) jointly endorse transactions, and blocks are committed only when a majority agrees. The result is an immutable, decentralized audit trail: as Prajay et al. observe, blockchain's "decentralized, transparent, and immutable" design lets users verify a pharmaceutical's traceability "from inception to conclusion".

Key Features: The platform addresses real-world supply-chain needs:

Unit-level Traceability: Each drug's serialized QR links to an on-chain asset. When registerDrug runs (at manufacture), it records fields like batch, expiration, and owner. Later transferDrug calls update the owner (e.g. "Central Hospital"). Because Hyperledger Fabric uses an append-only ledger, every handoff is recorded. In effect, scanning a product "tracks [it] from manufacture to patient".

Authentication: Scanning a QR runs our verifyDrug function. Instead of querying a centralized database, the code checks the shared ledger: if the serial exists and its status is "ACTIVE," the drug is genuine. This mirrors the regulated model where each saleable unit has a unique code checked against a database, but with decentralization. Fabric's permissioned design means no single company controls all data, yet all see consistent records. This greatly reduces counterfeit risk – studies show that

blockchain-backed systems make drug histories “impervious to manipulation”.

Auditability and Transparency: Every chaincode transaction (register, transfer, etc.) is automatically time-stamped and cryptographically signed by the submitting organization. Regulators or auditors can query getDrugHistory(serial) to retrieve a full chronicle of that drug’s journey. This matches best practice: as one survey notes, storing serialization on blockchain gives partners “an immutable audit trail”, something not possible with siloed databases.

Privacy and Compliance: We use Fabric’s MSP identities and TLS so only authorized peers (e.g. Org1=manufacturer, Org2=hospital) can read or write. Sensitive fields (like batch or quantity) are visible only on-chain, preventing any single party from altering them undetected. The system still meets global standards (US DSCSA, EU FMD, India’s track-and-trace) by encoding GTIN, serial, lot, and expiry in each code, but it removes the single point of failure in traditional central databases.

Implementation Details: We wrote the chaincode in JavaScript using Fabric’s contract API. The core smart-contract functions include:

registerDrug(serial, batch, name, manu, mfgDate, expDate, qty): Creates a new drug asset (status=“ACTIVE”).

readDrug(serial): Returns the current state (owner, status, etc.).

transferDrug(serial, newOwner): Updates the currentOwner field and logs the change (e.g. manufacturer → distributor → pharmacy).

verifyDrug(serial): Checks if the serial exists and returns true/false for authenticity.

getDrugHistory(serial): Returns a list of past owners, timestamps, and status changes for that serial number.

In testing on a two-peer network, we demonstrated key scenarios. For example, registering MED-000001 created an on-chain record for a Paracetamol batch from “ABC Pharma”. Querying immediately after returned the exact JSON payload we input (serial, batch, name, owner=“ABC Pharma”, status=“ACTIVE”). Transferring it to “Central Hospital” produced a successful commit (status:200) and updated the owner field. A subsequent verifyDrug showed that this package is valid and

current owner is now “Central Hospital.” Finally, getDrugHistory listed both the register and transfer transactions in order. This “closed-loop” test proves our system can secure the lifecycle of a drug unit, analogous to FLORA’s disease case but with ledger entries.

Comparison with Related Work: Our approach aligns with recent research in pharma blockchain. Nasreen et al. (2024) implemented a Hyperledger Fabric prototype “that stores the records of medicines on the blockchain” to improve traceability. Similarly, Prajay et al. survey multiple studies showing blockchain’s benefits for drug safety. These works emphasize the same core idea: a decentralized ledger replaces a vulnerable central system. Unlike pure academic prototypes, our solution is coupled with practical business logic (QR scanning, multi-org endorsement). We also drew inspiration from the FLORA project’s emphasis on connecting advanced tech with user workflows – here realized as a farmer (pharmacist)-oriented interface. For instance, just as FLORA uses Grad-CAM for explainability, we ensure QR-based feedback is straightforward for users: a green check for an authentic drug or a warning if something’s amiss.

User Interface and Platform: Though the focus has been backend, the end-user experience is critical. Our platform would present a simple UI (web or mobile) where users log in with their role (manufacturer, pharmacy, regulator). A user scans a package’s QR and immediately sees the drug’s details and history on screen, along with a green “Verified” or red “Alert”. We also envisioned features like multilingual support and integration with drug databases, so that by clicking on a serial a pharmacist can instantly view drug monographs or recall notices. Notifications (e.g. via email/SMS) could alert stakeholders if a transferred drug is scanned at the wrong location or appears on an unexpected timeline. These user-centric features mirror the “multilingual, notification” aspects FLORA highlighted, adapted to pharmaceutical compliance and ease-of-use.

Limitations and Future Work: Our current implementation is a proof-of-concept. It lacks features like off-chain storage or IoT integration. In the next phase, we would explore private data collections to encrypt commercially sensitive attributes (like cost or buyer identity), and integrating sensors (temperature, GPS) into each block for cold-chain validation. We also note that while blockchain greatly raises the bar against counterfeiters, it isn’t a silver bullet: attackers might still clone a valid QR if they replicate the physical label. Advanced measures like non-cloneable QR codes – which bind a code to a unique physical fingerprint – could be layered on later. Moreover, governance and performance need attention: scaling to dozens of organizations or handling thousands of scans per minute will require careful network tuning and perhaps hierarchical channels.

Conclusion: In summary, our blockchain-based tracking system addresses core pain points of drug supply chains: it enforces unit-level serialization and immutably logs every transaction, making it easy to detect fraud or errors. By replacing a central database with a shared ledger, we achieve the transparency and trust that the industry desperately needs. Early tests (registering, transferring, verifying a drug) confirm that each drug is securely recorded and traceable through Fabric's consensus model. As cited studies suggest, this strategy can significantly reduce counterfeit distribution and improve patient safety. Going forward, enhancing the system's interface and security will bring us closer to a deployable platform. Ultimately, this project demonstrates how distributed ledger technology – used thoughtfully – can bridge the gap between advanced IT and everyday pharmaceutical workflows, enabling more reliable drug authentication and recall-capability across India's healthcare network.

Sources: We built on established research in blockchain traceability and QR-based serialization, adapting these insights to our chaincode and platform design.

1.2 Existing Systems in Drug Supply Chain Tracking

Current drug distribution and tracking solutions exhibit significant limitations in transparency, security, and end-to-end visibility. Many systems rely on centralized databases or enterprise resource platforms where batch and serial data are stored in one place. While such systems simplify data aggregation, they create a single point of failure and trust bottleneck – stakeholders must trust the central authority and all raw data are pooled together. Other approaches remain largely manual or siloed (paper logs, spreadsheets, local databases). These are error-prone and offer virtually no immutable audit trail; as one review notes, "the prevailing lack of transparency in the existing system presents formidable hurdles" for verifying authenticity. Recently, some pilot projects employ basic blockchain ledgers: smart contracts record each transaction on a shared ledger. This inherently provides an immutable, tamper-evident record of drug movements. However, many of these early blockchain solutions are proofs-of-concept lacking advanced features – for example, they may omit fine-grained privacy controls or user-friendly interfaces. In summary, no single existing system fully meets all requirements of a modern drug-tracking platform: secure decentralization, privacy, scalability, and usability remain unmet in the current landscape.

Key existing approaches include:

Centralized Database Systems: Drug manufacturers and distributors often use centralized

servers or ERP systems to log production and shipping events. These systems can efficiently process serialized data, but as studies note, centralized storage creates a single point of failure and data-sharing concerns. Privacy is limited since all data reside together; regulators and partners must rely on the database owner for audit access.

Traditional Manual Tracking: Smaller actors or legacy systems may still rely on paper records, spreadsheets, or isolated digital files. Such methods are highly error-prone and offer virtually no real-time visibility. As one analysis observes, "lack of transparency in the existing system makes it exceedingly difficult to verify product authenticity or detect tampering". These manual approaches also scale poorly for nationwide or multi-entity traceability.

Basic Blockchain Systems: Several experimental blockchain solutions have been proposed for drug tracing. In these, each handoff is recorded on a distributed ledger, yielding an immutable chain-of-custody. For example, researchers describe a "blockchain-based Drug ledger" designed to address security and efficiency shortcomings in traditional systems. However, many current blockchain pilots lack full integration: they may not implement robust privacy layers, consensus policies across organizations, or user-facing apps for non-technical stakeholders. Interoperability and standardization across diverse industry participants remain open challenges.

The table below compares these existing approaches against our proposed blockchain-based solution. It highlights how each parameter is handled differently in centralized, manual, and basic blockchain systems, versus our design.

Table 1.1: Existing System vs Proposed System

Parameter	Centralized Database Systems	Traditional Tracking Systems	Basic Blockchain Systems	DrugTrace (Proposed System)
Input Data Source	Scanned barcodes/RFID fed into a central server	Manual logs or standalone barcode scans	Local scans with periodic ledger sync	Smartphone/NFC QR-code scans with instant blockchain entry

Data Handling	Central SQL/ERP database storing all records	Siloed entry (paper or local DB)	Distributed ledger (each node holds copy)	Permissioned ledger (Fabric) with integration APIs
Tracking Mechanism	Central database queries by batch/serial number	Visual/manual inventory checks	Smart-contract records on each transfer	Chaincode-driven register/transfer with in-app verification
Data Distribution	Single central repository	No shared data (fully isolated)	Replicated across all blockchain nodes	Distributed ledger across all stakeholders (permissioned)
Privacy	Minimal (all raw data centralized)	Variable (physical security only)	Permissioned view; sensitive fields hashed or omitted	Encrypted data on-chain; fine-grained access control
Consensus Mechanism	Central authority validates	No consensus (trust between parties)	Standard permissioned consensus (RAFT/IBFT)	Multi-org endorsement policies (Hyperledger Fabric)
Traceability	Basic audit logs in ERP; paper trail	Hard to trace events	Immutable chain-of-custody on ledger	Queryable provenance; tamper-evident audit trail

Real-World Challenges	Single point-of-failure; downtime impacts all users	Error-prone; delayed/manual updates	Adoption and interoperability issues	Supports offline scanning; designed for resilience
Output	Centralized batch/shipment reports	Printed logbooks	On-chain transaction records	Digital certificates; blockchain-verified data outputs
Product Information	Full drug details stored in ERP	Only basic label info available	Hashes or references to metadata	Complete drug metadata on-chain or linked off-chain
Transaction History	Centralized audit logs	Separate logs at each entity	Full history replicated on blockchain	Immutable history accessible via user interface
User Accessibility	Complex enterprise dashboards	Office staff only	Developer/CLI tools	User-friendly web/mobile apps (pharmacists, regulators)
Multilingual Support	Typically one default language	Local language via forms	Usually English	Supports multiple languages in UI
Regulatory Compliance	Manual compliance reporting	Often weak or ad-hoc	Immutable records help audits	Built-in support for DSCSA/GMP standards

Alerts/Notifications	Rare (e.g. email alerts)	None	None	Automated recall/expiration and inventory alerts
Backend	On-premises or cloud ERP (SQL databases)	Paper or local files/databases	Permissioned blockchain (Fabric/Ethereum)	Hyperledger Fabric network + Node.js chaincode + web/mobile UI
Scalability	Scales with added server capacity	Limited by manual processing	Scales by adding nodes (subject to consensus overhead)	Designed for multi-organization scaling
Main Limitation	Single point of failure; heavy trust required	Error-prone; not scalable	Complex deployment; governance issues	Requires broad ecosystem adoption; emerging technology

1.3 PROBLEM STATEMENT

The pharmaceutical sector is an environment where accurate inventory and reliable supply-chain tracking can make a significant difference — better availability, reduced wastage, and improved patient safety. The challenge is that traditional drug inventory systems often depend on separate databases, manual records, and information shared between manufacturers, distributors, warehouses, hospitals, pharmacies, and regulatory organizations. This creates difficulties in maintaining a consistent and trustworthy view of drug movement across multiple stakeholders. This project takes a different approach: a blockchain-based Drug Inventory and Supply Chain Tracking System that provides secure and auditable tracking of medicines across their supply-chain journey. Drug records can be registered with unique identifiers, while ownership and movement information can be recorded through blockchain transactions. The system uses Hyperledger Fabric to maintain trusted supply-chain records,

while application-level services can manage inventory, alerts, verification, and operational information.

The system is designed to track important drug information such as serial numbers, batch numbers, manufacturer details, manufacturing and expiry dates, quantity, current ownership, and status. A QR-based verification mechanism can provide convenient access to the digital drug record, while the blockchain maintains the trusted history of registered drugs and ownership transfers. However, a QR code by itself cannot prove that a physical medicine is genuine, since a genuine identifier can potentially be copied. Stronger anti-counterfeit protection therefore requires unique serialization, scan-history analysis, duplicate-scan or location anomalies, tamper-evident packaging, and potentially cryptographic, NFC, or RFID-based mechanisms.

1.4 The proposed system is built around four core problems:

1.4.1 Delayed and Difficult Drug Traceability:

Drug supply-chain monitoring has traditionally relied on manual records, invoices, spreadsheets, and separate inventory systems maintained by different organizations. For manufacturers, distributors, warehouses, and hospitals, obtaining an accurate view of where a particular drug is located, who currently owns it, and how it reached its present location can be difficult. Delayed updates, incomplete records, and human errors can make it harder to identify stock movement, expired medicines, shipment delays, or irregular transactions in time.

1.4.2 Fragmented Data Sharing and Trust Concerns:

Standard pharmaceutical supply-chain processes often require drug and inventory information to be exchanged between multiple independent organizations. Each participant may maintain its own database or documentation, resulting in fragmented records and limited end-to-end visibility. Centralized systems also introduce concerns related to data ownership, security, privacy, unauthorized modification, and dependence on a single controlling authority. Organizations may be reluctant to share sensitive operational or commercial information directly, making reliable coordination across the complete supply chain more difficult.

1.4.3. Inventory, Ownership, and Counterfeit Management Challenges:

Pharmaceutical organizations must continuously manage stock levels, batch information,

expiry dates, ownership transfers, and shipment movements. Manual monitoring can result in stockouts, overstocking, expired inventory, delayed procurement, and difficulties during audits or recalls. Counterfeit or unauthorized medicines create an additional challenge because conventional barcode or QR verification may only confirm that an identifier exists in a database. A reliable system must connect unique drug identities with a tamper-evident history of registration, transfers, verification events, and anomaly detection so that suspicious activity can be identified more effectively.

1.4.4. Limited Transparency and Auditability of Drug Transactions:

Many existing systems provide only limited visibility into the complete history of a medicine across different supply-chain stages. When records are maintained independently, reconstructing the complete sequence of registration, shipment, ownership transfer, and verification can become time-consuming. A practical pharmaceutical supply-chain system needs a trusted and auditable transaction history that authorized stakeholders can use to verify drug information, monitor movement, support regulatory audits, and investigate suspicious or inconsistent activities. Hyperledger Fabric provides a permissioned distributed-ledger foundation for maintaining such critical records without relying on a public blockchain.

1.5 PROPOSED SYSTEM

The proposed Drug Inventory and Supply Chain Tracking System is designed as an end-to-end pharmaceutical supply-chain platform that brings together blockchain technology, smart contracts, drug serialization, inventory management, verification, shipment tracking, and stakeholder-oriented digital services under one system. Authorized users such as manufacturers, distributors, warehouses, hospitals, and other supply-chain participants can interact with the platform to register drugs, view inventory information, verify drug records, track ownership, and monitor supply-chain transactions. Each drug can be associated with a unique serial number containing important information such as batch number, drug name, manufacturer, manufacturing date, expiry date, quantity, current owner, and status. A QR-based interface can provide convenient access to the corresponding digital drug record, while the blockchain maintains the trusted transaction history. In the current prototype, Hyperledger Fabric v2.5.16 is used with Org1 representing the Manufacturer and Org2 representing the Hospital, connected through the drugchannel channel and the drugtrace chaincode.

At the heart of the system is the permissioned blockchain network itself. Rather than depending entirely on a single centralized database, authorized organizations can maintain and verify a

shared record of important drug transactions. The drugtrace smart contract currently provides functions such as registerDrug, readDrug, drugExists, transferDrug, verifyDrug, and getDrugHistory. When a drug is registered, its identity and relevant information are recorded on the ledger. Ownership can subsequently be transferred between authorized supply-chain participants, while verification can be performed against the recorded drug identity and status. The current prototype demonstrates this workflow using the sample drug MED-000001. The proposed architecture can later be extended to include distributors, warehouses, pharmacies, regulators, and additional organizational participants. Node.js is intended to act as the application API and backend layer, while PostgreSQL can maintain off-chain operational information that does not need to be stored directly on the blockchain.

1.6 BENEFITS OF THE PROJECT

The proposed Drug Inventory and Supply Chain Tracking System provides significant benefits across pharmaceutical, technological, security, operational, and healthcare dimensions:

1.6.1. Secure and Transparent Drug Traceability

By using Hyperledger Fabric, the proposed system enables authorized pharmaceutical stakeholders to maintain a trusted record of important drug transactions without depending entirely on a single centralized record. Drug registration, ownership transfers, verification, and historical transactions can be recorded on the permissioned ledger, improving transparency and traceability across different stages of the supply chain.

1.6.2. Improved Drug Verification and Anti-Counterfeit Support

The system associates drugs with unique serialized identities and provides QR-assisted verification of their corresponding digital records. This allows authorized users to check whether a drug is registered, identify its batch and manufacturer, and view its current status and ownership. QR is not treated as proof of physical authenticity by itself; stronger protection can be achieved through secure serialization, duplicate-scan detection, scan-history analysis, tamper-evident packaging, and future cryptographic, NFC, or RFID mechanisms.

1.6.3. Accurate Inventory and Ownership Management

Recording drug information such as serial number, batch number, quantity, manufacturing date, expiry date, current owner, and status provides a structured approach to pharmaceutical inventory

management. Ownership transfers can be recorded through smart-contract transactions, helping manufacturers, distributors, warehouses, and hospitals maintain a clearer view of drug movement and reduce errors associated with manual record keeping.

1.6.4. Improved Auditability and Data Integrity

Blockchain-based transaction records provide a tamper-evident history of important drug supply-chain events. The system can retrieve the historical transactions associated with a drug, making it easier for authorized stakeholders to investigate ownership changes, verification activities, and other supply-chain events. This can reduce the effort required for manual reconciliation and support regulatory audits and compliance activities.

1.6.5. Integrated Supply-Chain Monitoring and Inventory Support

Instead of focusing only on drug verification, the proposed system is designed to combine inventory management, shipment and ownership tracking, stock and expiry alerts, notifications, dashboards, and reporting within a unified platform. This can help organizations identify low-stock conditions, monitor approaching expiry dates, track drug movement, and make more informed procurement and distribution decisions.

1.6.6. Scalable and Accessible Pharmaceutical Supply-Chain Platform

By combining a permissioned Hyperledger Fabric network with an application backend, off-chain operational storage, and Android/web interfaces, the proposed architecture can be extended to additional stakeholders such as distributors, warehouses, pharmacies, and regulatory organizations. Future enhancements such as AI-based demand forecasting, procurement optimization, advanced anomaly detection, IoT-based storage monitoring, and automated recall management can further improve the scalability, intelligence, and effectiveness of the pharmaceutical supply-chain system.

CHAPTER 2
LITERATURE SURVEY

2.1 INTRODUCTION

A comprehensive literature survey is essential to establish the scientific and technological foundation for the proposed Drug Inventory and Supply Chain Tracking System. The rapid convergence of blockchain technology, pharmaceutical information systems, supply-chain management, drug serialization, secure verification, inventory analytics, and digital healthcare has resulted in extensive research on improving the safety, transparency, and efficiency of pharmaceutical distribution. However, existing academic studies and healthcare applications frequently address these areas as separate components rather than integrating them into a unified, secure, traceable, and stakeholder-oriented platform.

To establish a strong theoretical and technical baseline, this literature survey examines research across six key domains: **(1)** pharmaceutical inventory management and digital drug supply-chain systems; **(2)** blockchain and Distributed Ledger Technology for secure and tamper-evident pharmaceutical traceability; **(3)** drug serialization, QR-based identification, and anti-counterfeit verification mechanisms; **(4)** smart-contract-based ownership and transaction management across multiple supply-chain participants; **(5)** inventory analytics, stock and expiry monitoring, and AI-based demand forecasting for pharmaceutical procurement; and **(6)** integrated healthcare and pharmaceutical platforms incorporating dashboards, shipment tracking, alerts, audit trails, and role-based access. A critical examination of these research areas highlights the need for an integrated system that combines reliable drug identification, permissioned blockchain-based traceability, inventory management, secure verification, and practical supply-chain support within a single scalable pharmaceutical framework.

2.2 RELATED WORK

2.2.1 Blockchain-Based Pharmaceutical Supply Chain Tracking

Existing research on blockchain-based pharmaceutical supply chains has investigated the use of Distributed Ledger Technology to improve transparency and traceability throughout the movement of medicines. Such approaches record important supply-chain events, including drug registration, manufacturing information, shipment activities, ownership changes, and delivery records, on a shared

ledger maintained by authorized participants. Permissioned blockchain platforms such as Hyperledger Fabric are particularly suitable for pharmaceutical environments because participating organizations can be identified and access to sensitive information can be controlled.

Key Limitation: Many blockchain-based pharmaceutical tracking approaches concentrate primarily on recording supply-chain transactions and improving ledger transparency. They may not provide a complete inventory-management layer, practical verification workflow, stock and expiry alerts, dashboards, or advanced anti-counterfeit mechanisms. The proposed system addresses this gap by combining permissioned blockchain traceability with drug inventory management, verification, ownership tracking, and supply-chain monitoring.

2.2.2 Centralized Pharmaceutical Inventory and Drug Tracking Systems

Existing centralized pharmaceutical information systems provide organizations with facilities for maintaining drug inventories, batch information, expiry dates, procurement records, and distribution information. Such systems can improve operational efficiency within an individual organization and provide centralized access to inventory data. However, when manufacturers, distributors, warehouses, hospitals, pharmacies, and regulators operate separate systems, maintaining a consistent end-to-end view of a drug becomes difficult. Data must frequently be exchanged between organizations, creating challenges related to interoperability, data ownership, security, and trust.

Key Limitation: Centralized approaches depend on a central database or controlling organization and may provide limited visibility across independently managed supply-chain participants. Unauthorized modification, delayed updates, fragmented records, and dependence on a single system can also affect auditability and reliability. The proposed system addresses these limitations through a permissioned blockchain network in which authorized organizations can maintain and verify a shared record of critical drug transactions.

2.2.3 Serialization and Digital Verification for Pharmaceutical Products

Research and practical pharmaceutical systems have increasingly explored product serialization and machine-readable identifiers such as barcodes and QR codes to improve medicine identification and traceability. A unique identifier can associate a physical medicine package with information such as its product identity, batch, manufacturer, expiry date, and supply-chain status. Digital verification allows authorized users to retrieve the corresponding product information and identify records that are missing, invalid, or inconsistent with the expected supply-chain history.

Key Limitation: A QR code or barcode alone cannot guarantee that the physical medicine is genuine because an identifier printed on a legitimate package can potentially be copied onto another package. Therefore, effective anti-counterfeit protection requires more than simple code scanning. Secure per-package serialization, blockchain-backed transaction history, duplicate-scan detection, unusual scan-location or timing analysis, tamper-evident packaging, and potentially cryptographic, NFC, or RFID mechanisms can provide stronger protection. The proposed system treats QR as a convenient identification and verification mechanism while using blockchain to protect the integrity of the associated digital record.

2.2.4 Smart Contracts for Drug Ownership and Supply-Chain Transactions

Blockchain-based pharmaceutical systems can use smart contracts to automate and validate important supply-chain operations. Smart contracts can represent business rules for drug registration, ownership transfer, verification, and status management, reducing dependence on manual reconciliation between participating organizations. Such transaction-based approaches can create a consistent history of drug movement and provide authorized stakeholders with a reliable record for auditing and investigation.

Key Limitation: Basic implementations may provide only transaction recording without incorporating broader inventory operations, user-facing dashboards, alerts, or analytical capabilities. They may also rely on trusted input data without independently proving the physical condition or authenticity of a medicine. The proposed system extends the smart-contract approach by combining drug registration, ownership transfer, verification, historical tracking, and planned inventory-management functions within a broader pharmaceutical supply-chain platform.

2.2.5 AI-Based Pharmaceutical Demand Forecasting and Inventory Optimization

Research in pharmaceutical inventory management has also explored statistical and machine-learning techniques for predicting medicine demand and improving procurement decisions. Demand forecasting can use historical consumption, inventory levels, seasonal patterns, disease trends, and distribution information to estimate future requirements. Such approaches can help organizations reduce stockouts, emergency procurement, overstocking, and medicine expiry while improving resource utilization.

Key Limitation: Forecasting systems are often developed as separate analytical components and may not be directly connected to a trusted, end-to-end drug traceability system. Predictions can also be

affected when inventory or consumption records are incomplete, delayed, or inconsistent. The proposed system is designed to provide a foundation where reliable inventory and supply-chain records can later support AI-based demand forecasting and procurement optimization as a future enhancement.

2.2.6 Integrated Pharmaceutical Supply-Chain Management Platforms

Recent digital healthcare and pharmaceutical-management approaches increasingly combine inventory monitoring, shipment management, product verification, analytics, and reporting into integrated platforms. Such systems aim to provide stakeholders with dashboards that summarize stock levels, drug movement, expiry information, procurement requirements, and supply-chain performance. Role-based access and notifications can further improve operational usability for different participants such as manufacturers, distributors, hospitals, and administrators.

Key Limitation: Many integrated platforms remain dependent on centralized databases and do not provide a permissioned, shared, tamper-evident transaction history across independent organizations. Others may provide blockchain traceability without sufficient inventory intelligence or user-oriented operational features. The proposed Drug Inventory and Supply Chain Tracking System aims to bridge these gaps by combining Hyperledger Fabric-based traceability, smart-contract transactions, serialized drug identities, QR-assisted verification, inventory management, shipment and ownership tracking, alerts, dashboards, and future AI-based demand forecasting within one unified platform.

2.3 INFERENCE FROM RELATED WORK:

A holistic synthesis of the surveyed literature identifies several important insights that define the technical architecture and development direction of the **Drug Inventory and Supply Chain Tracking System**:

2.3.1. Permissioned Blockchain is Important for Multi-Organization Drug Traceability

Existing centralized pharmaceutical supply-chain systems generally require drug and inventory information to be maintained by a central organization or exchanged between separate organizational databases. Permissioned Blockchain provides an alternative in which authorized participants can maintain and verify a shared record of critical supply-chain transactions. This establishes secure, auditable, and multi-organization traceability as an important component of the proposed system.

2.3.2. Unique Drug Serialization and Digital Verification are Essential for Pharmaceutical Tracking

Existing drug-tracking approaches demonstrate the importance of identifying pharmaceutical products using unique product or package identifiers. The proposed system therefore associates drug records with unique serial numbers containing information such as batch number, drug name, manufacturer, manufacturing date, expiry date, quantity, current owner, and status. QR codes can provide convenient access to these digital records; however, QR verification alone cannot guarantee physical authenticity. Stronger anti-counterfeit protection can be achieved through secure serialization, scan-history monitoring, duplicate-scan detection, tamper-evident packaging, and potentially cryptographic, NFC, or RFID mechanisms.

2.3.3. Smart Contracts are Suitable for Managing Drug Transactions and Ownership

Pharmaceutical supply chains involve multiple transactions such as drug registration, shipment, ownership transfer, verification, and historical tracking. Smart contracts can provide structured rules for recording these activities and reduce dependence on manual reconciliation between participants. The proposed system therefore uses the **drugtrace** chaincode to implement functions such as registerDrug, readDrug, drugExists, transferDrug, verifyDrug, and getDrugHistory, establishing a foundation for automated and auditable drug-supply-chain transactions.

2.3.4. Data Integrity and Auditability are Critical for Pharmaceutical Supply Chains

Drug records must remain reliable throughout their lifecycle because inaccurate or modified information can affect inventory decisions, regulatory audits, recalls, and patient safety. Blockchain-based transaction histories provide a tamper-evident record of important drug events and allow authorized participants to review the history associated with a particular drug. The proposed system therefore incorporates Hyperledger Fabric as the trusted ledger layer for maintaining drug registration, ownership-transfer, verification, and historical transaction records.

2.3.5. Drug Traceability Should be Integrated with Inventory and Supply-Chain Support:

The surveyed approaches primarily concentrate on individual aspects such as drug traceability, inventory management, or counterfeit verification. This indicates scope for extending the blockchain tracking process into a complete pharmaceutical platform that combines drug inventory,

shipment and ownership tracking, batch and expiry monitoring, stock alerts, verification, dashboards, reporting, and notifications. The proposed system aims to integrate these capabilities so that stakeholders can monitor the movement and availability of medicines through a unified platform.

2.3.6. Scalable and Intelligent Pharmaceutical Management Can Improve Practical Applicability

Existing pharmaceutical systems can be extended beyond basic transaction tracking by incorporating application interfaces, operational databases, analytics, and intelligent decision-support mechanisms. The proposed architecture therefore considers Android and web interfaces for practical access, Node.js as the application API/backend layer, and PostgreSQL for off-chain operational information. Future enhancements can incorporate AI-based demand forecasting, procurement optimization, predictive expiry management, anomaly detection, IoT-based storage monitoring, and automated recall support, providing a foundation for a scalable and intelligent pharmaceutical supply-chain ecosystem.

CHAPTER 3
SYSTEM DESIGN

3.1 INTRODUCTION

System design establishes the architectural blueprint of the **Drug Inventory and Supply Chain Tracking System**, defining its functional components, communication interfaces, data flow, processing pipeline, and security boundaries. The overall design emphasizes modularity, scalability, data integrity, traceability, and controlled integration between pharmaceutical stakeholders and the blockchain-based supply-chain components. The architecture combines an Android/web-based user interface with a Node.js application and API layer, PostgreSQL for off-chain operational data, and Hyperledger Fabric for maintaining trusted drug and supply-chain transactions. Smart contracts manage important operations such as drug registration, ownership transfer, verification, and historical tracking, while application-level services support inventory management, alerts, shipment monitoring, dashboards, and reporting.

The proposed system is designed to support the different requirements involved in a pharmaceutical supply-chain environment, including **(1)** manufacturers requiring secure drug registration, batch management, inventory visibility, and supply-chain tracking; **(2)** distributors, warehouses, and hospitals requiring verification, ownership transfer, shipment information, and stock monitoring; and **(3)** administrators and authorized stakeholders requiring controlled management of users, drug records, inventory information, alerts, reports, and audit history. To satisfy these requirements, the system follows a modular architecture consisting of the User and Presentation Module, Drug and Inventory Management Module, Verification and Serialization Module, Blockchain and Smart Contract Module, Backend and API Module, and Persistence, Analytics, and Reporting Module. These components communicate through defined interfaces to provide an integrated, secure, and auditable pharmaceutical supply-chain tracking system.

3.2 SYSTEM ARCHITECTURE

The Drug Inventory and Supply Chain Tracking System is organized into six functional layers, each fulfilling a distinct operational objective in the proposed pharmaceutical platform. The architecture connects manufacturers, distributors, warehouses, hospitals, pharmacies, and administrators with the application interface, backend services, blockchain network, inventory services, and persistent data storage. The design emphasizes modularity, permissioned access, secure drug traceability, reliable inventory management, and integration of blockchain-based transaction records with off-chain

operational information.

3.2.1 User and Presentation Layer

This layer provides the primary interaction between the system and its authorized users, including manufacturers, distributors, warehouse personnel, hospital staff, pharmacies, and administrators. The frontend is intended to provide facilities such as user authentication, dashboard access, drug registration, inventory viewing, QR-based verification, shipment tracking, ownership information, alerts, and reporting. Users can interact with the platform to register or identify drugs, view relevant drug information, verify recorded details, monitor inventory, and review supply-chain transactions. The presentation layer acts as the main entry point through which authorized users communicate with the backend, inventory services, and blockchain components.

3.2.2 Drug Identification and Verification Layer

This layer manages the identification and verification of pharmaceutical products throughout the supply chain. Each drug record can contain a unique serial number, batch number, drug name, manufacturer, manufacturing date, expiry date, quantity, current owner, and status. QR codes can be associated with serialized drug records to provide convenient access to their digital information. The verification process checks the corresponding blockchain record to determine whether the drug is registered and whether its recorded information is valid. The system does not treat a QR code alone as proof of physical authenticity; stronger anti-counterfeit capabilities can be added through secure serialization, duplicate-scan detection, scan-history analysis, tamper-evident packaging, and cryptographic, NFC, or RFID technologies.

3.2.3 Backend and API Layer

The backend layer is intended to be implemented using Node.js and acts as the central communication and business-logic layer of the system. It provides REST APIs, authentication and authorization, business logic, blockchain communication, inventory operations, and application-level request handling. The backend manages communication between the user interface, PostgreSQL database, Hyperledger Fabric network, and other system services. It can coordinate requests related to users, drug records, inventory, verification, shipment information, notifications, and reporting while ensuring that access to sensitive operations is restricted to authorized participants.

3.2.4 Blockchain and Smart Contract Layer

This layer forms the core trusted transaction component of the proposed system. The architecture uses Hyperledger Fabric as a permissioned distributed ledger in which authorized organizations can participate in pharmaceutical supply-chain transactions. The current prototype uses Hyperledger Fabric v2.5.16, with Org1 representing the Manufacturer and Org2 representing the Hospital, connected through the drugchannel channel. The drugtrace chaincode provides functions including registerDrug, readDrug, drugExists, transferDrug, verifyDrug, and getDrugHistory. These functions provide the foundation for registering drugs, checking their existence, transferring ownership, verifying records, and retrieving historical transactions. The current prototype has successfully demonstrated drug registration using the sample serial number MED-000001.

3.2.5 Inventory, Shipment, and Supply-Chain Management Layer

This layer manages the operational information associated with pharmaceutical inventory and movement. It is designed to provide information regarding stock quantities, batch details, expiry dates, current ownership, shipment status, and supply-chain events. Manufacturers and other authorized participants can monitor drug availability and movement between supply-chain stages, while hospitals and other receiving organizations can verify received products against their recorded information. The layer can also support stock-level alerts, expiry notifications, shipment monitoring, recall-related activities, and emergency inventory transfers. AI-based demand forecasting and procurement optimization can be incorporated into this layer as future enhancements.

3.2.6 Persistence, Analytics, and Reporting Layer

The system uses **PostgreSQL** as the planned off-chain database for operational information that does not need to be stored directly on the blockchain. This layer can maintain information such as **user profiles, inventory records, notifications, scan logs, shipment details, analytics data, and application-level records**, while critical drug identity and transaction events are maintained through Hyperledger Fabric. Dashboards and reports can present inventory status, drug verification results, shipment and ownership information, expiry conditions, and supply-chain activity to authorized users. This separation between on-chain and off-chain information helps keep the blockchain focused on trusted traceability events while allowing the application to efficiently manage operational and analytical data.

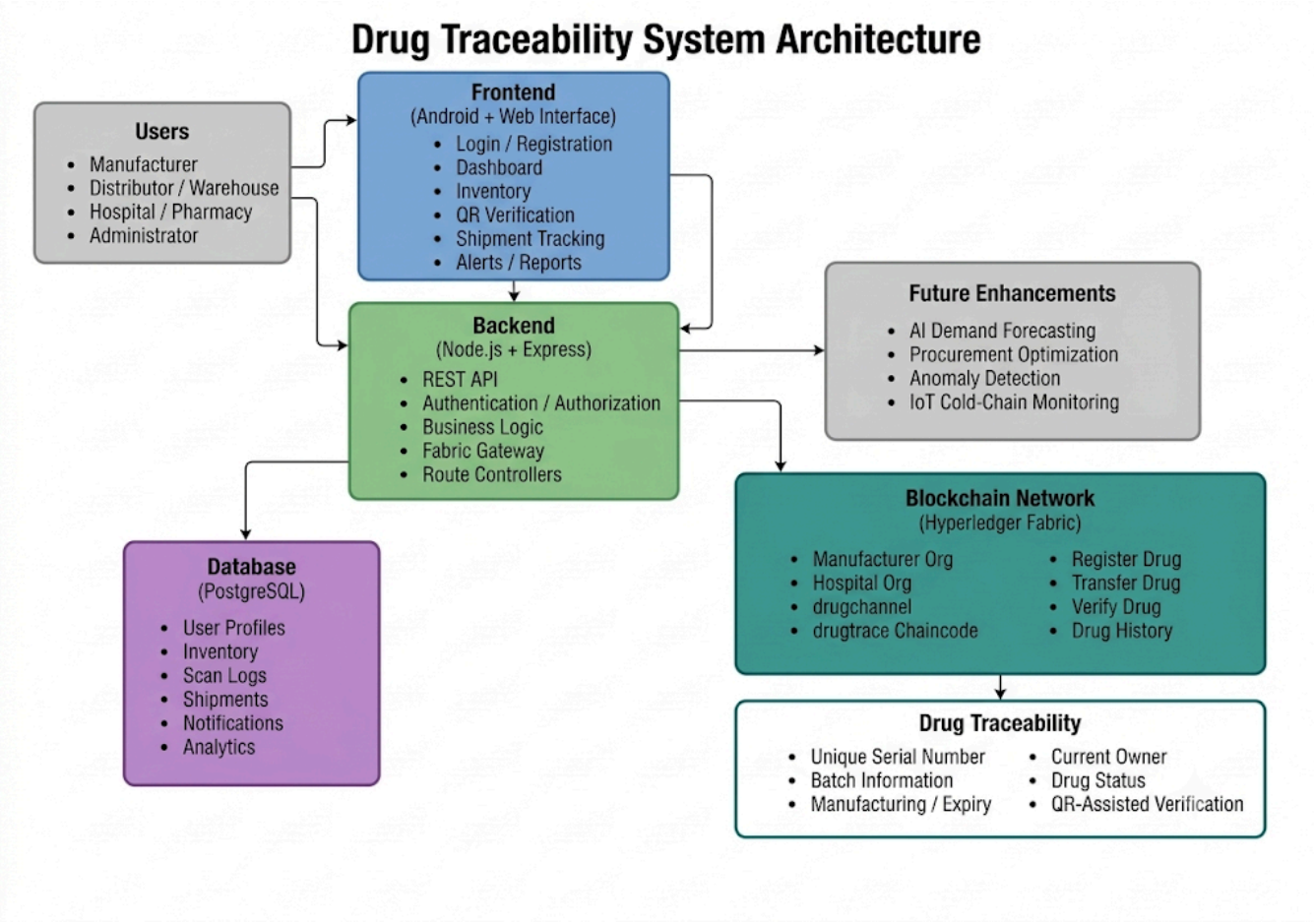


Figure 3.1:

System Architecture Diagram of drug traceability Showing User Interface, frontend, Backend, future enhancements, blockchain network, database, drug traceability.

3.3 SYSTEM REQUIREMENTS

The technology stack for the **Drug Inventory and Supply Chain Tracking System** is selected to support secure pharmaceutical inventory management, blockchain-based drug traceability, supply-chain transaction processing, QR-assisted verification, and stakeholder-oriented application access. The architecture combines an Android/web-based user interface with a Node.js backend and API layer, PostgreSQL for off-chain operational data, and Hyperledger Fabric for maintaining trusted drug and supply-chain transactions. Smart contracts implemented through Fabric chaincode manage important operations such as drug registration, drug verification, ownership transfer, and transaction history. The proposed system can additionally support inventory dashboards, shipment tracking, stock and expiry alerts, reporting, and future AI-based demand forecasting.

Table 3.1: Software Tools and Technology Stack

Component	Technology / Version	Purpose
Mobile Application	Android / Kotlin	Mobile interface for drug verification, inventory access and supply-chain operations
Web Application	Web-based Interface	Dashboard and management interface for authorized pharmaceutical stakeholders
UI Framework	Android UI / Web UI	Provides user-friendly screens for inventory, verification, shipment and reporting operations
Backend / API Layer	Node.js + Express	REST APIs, business logic, request handling and communication with blockchain services
Blockchain Platform	Hyperledger Fabric v2.5.16	Permissioned distributed ledger for trusted drug and supply-chain transactions
Blockchain Network	Fabric Test Network	Provides the development environment for participating pharmaceutical organizations
Organizations	Org1 Manufacturer / Org2 Hospital	Represents participating organizations in the current blockchain prototype

Blockchain Channel	drugchannel	Provides a communication and ledger channel for the participating organizations
Smart Contract / Chaincode	Node.js JavaScript – drugtrace	Implements drug registration, verification, ownership transfer and history functions
Chaincode Functions	registerDrug, readDrug, drugExists, transferDrug, verifyDrug, getDrugHistory	Performs core drug-tracking and verification operations
Database	PostgreSQL	Planned off-chain storage for users, inventory, scan logs, shipments, notifications and analytics
Fabric Gateway	Hyperledger Fabric Gateway	Intended backend interface for communication between the application and Fabric network
Drug Identification	Unique Serial Number	Provides a unique digital identity for individual drug records
QR Verification	QR-based Drug Identification	Provides convenient access to the corresponding digital drug record
Inventory Management	Application-level Inventory Module	Manages stock quantity, batch information, expiry details and inventory status
Shipment / Ownership Tracking	Blockchain Transactions + Backend Services	Tracks drug movement and ownership changes between supply-chain participants

Notifications and Alerts	Application Notification Module	Provides stock, expiry, shipment and other supply-chain alerts
Reporting and Dashboard	Web / Android Dashboard	Displays inventory, verification, shipment and supply-chain information
AI Demand Forecasting	Machine Learning – Future Enhancement	Supports future demand prediction and procurement optimization
Development Environment	Visual Studio Code	Primary environment for backend, chaincode and application development
Containerization	Docker	Runs Hyperledger Fabric network components and supporting services
Version Control	Git + GitHub	Source-code management, collaboration and project version tracking

Functional Requirements Specification:

The functional requirements define the mandatory operations, processing activities, and system workflows that the Drug Inventory and Supply Chain Tracking System should perform to provide secure drug traceability, inventory monitoring, verification, and pharmaceutical supply-chain management.

Table 3.2: Functional Requirements Specification

ID	Functional Requirement	Description

FR-01	User Registration and Authentication	The system shall allow authorized users to register and securely access the application according to their assigned role and organization.
FR-02	Role-Based Access	The system shall provide appropriate access permissions for manufacturers, distributors, warehouse personnel, hospitals, pharmacies and administrators according to their responsibilities.
FR-03	Drug Registration	The system shall allow authorized manufacturers to register drug records containing serial number, batch number, drug name, manufacturer, manufacturing date, expiry date, quantity, owner and status.
FR-04	Drug Record Management	The system shall allow authorized users to retrieve and view registered drug information using the corresponding serial number or supported identification mechanism.
FR-05	Drug Verification	The system shall verify whether a requested drug record exists on the blockchain and shall display relevant information such as drug name, batch number, manufacturer, expiry date, current owner and status.
FR-06	QR-Assisted Verification	The system shall support QR-based identification of a drug record so that authorized users can conveniently access its digital supply-chain information.
FR-07	Ownership Transfer	The system shall record the transfer of drug ownership between authorized supply-chain participants and maintain the updated current-owner information.
FR-08	Drug History Tracking	The system shall maintain and provide access to the historical transactions associated with a drug, including registration and ownership-transfer events.
FR-09	Inventory Management	The system shall maintain inventory information including drug quantity, batch details, expiry dates, ownership and inventory status for authorized organizations.

FR-10	Shipment Tracking	The system shall support recording and monitoring of drug movement and shipment-related information between participating supply-chain organizations.
FR-11	Stock and Expiry Alerts	The system shall provide alerts for low-stock conditions, approaching expiry dates and other important inventory conditions.
FR-12	Supply-Chain Dashboard	The system shall provide dashboards displaying relevant inventory, drug verification, shipment, ownership and supply-chain information to authorized users.
FR-13	Audit and Reporting	The system shall provide transaction history and reports that assist authorized users in auditing drug movement, verification activities and supply-chain operations.
FR-14	Administrative Management	Authorized administrators shall be able to manage users, organizations, system information and access permissions while maintaining appropriate security restrictions.
FR-15	Secure Data Management	The system shall protect sensitive application information through authenticated access, role-based authorization and appropriate separation of blockchain and off-chain operational data.

Hardware Requirements Specification:

The hardware requirements are designed to support application development, machine-learning experimentation, federated-learning simulation, testing, and deployment without requiring specialized proprietary agricultural hardware. Since EfficientNetB0 is comparatively lightweight, the application and model-development environment can operate on a modern development computer, while computationally intensive federated experiments may optionally use GPU-enabled systems or cloud environments.

Table 3.3: Hardware Requirements Specification

Hardware Component	Minimum Requirement	Recommended Requirement	Purpose
Development CPU	Intel Core i5 / AMD Ryzen 5, 4 cores	Intel Core i7 / AMD Ryzen 7 or equivalent	Backend development, blockchain execution, database operations and application development
RAM	8 GB	16 GB or higher	Running Fabric containers, Docker, development tools, database and application services
Storage	256 GB SSD	512 GB NVMe SSD or higher	Source code, Docker images, blockchain data, database files and project resources
GPU	Integrated GPU / Not Required	Dedicated GPU	Optional acceleration for future AI demand forecasting and analytics
Display	1366 × 768	1920 × 1080 Full HD	Application development, dashboard testing

			and monitoring
Network Connection	5 Mbps broadband	25 Mbps or higher	Communication between application services, blockchain organizations and deployment environments
Android Test Device	Android smartphone	Android 11 or later	Testing mobile application, QR verification and supply-chain workflows
Camera	5 MP or above	12 MP or above	Scanning or capturing QR codes and supporting future package-identification workflows
Server / Cloud Resources	Basic cloud or local server	Scalable multi-service server	Hosting backend, PostgreSQL, blockchain components and application services
Docker Environment	Docker-compatible system	Docker with sufficient CPU/RAM resources	Running Hyperledger Fabric peers, orderer and supporting blockchain services

3.4 DATA FLOW DIAGRAM / USE CASE DIAGRAM

Data Flow Architecture (Level-0 and Level-1)

The flow of information through the Drug Inventory and Supply Chain Tracking System is designed as a modular pipeline connecting pharmaceutical stakeholders, application services, blockchain components, inventory management, and persistent storage. At the user touchpoint, manufacturers, distributors, warehouse personnel, hospitals, pharmacies, and administrators interact with the Android/web-based application to register drugs, view inventory, verify drug information, monitor shipments, manage ownership transfers, and review supply-chain records. Drug information submitted through the application is transferred to the backend processing layer, where it is validated and routed to the appropriate inventory, database, or blockchain service.

The drug registration and verification workflow uses unique drug identifiers containing information such as serial number, batch number, drug name, manufacturer, manufacturing date, expiry date, quantity, current owner, and status. When a registered drug is verified, the backend communicates with the Hyperledger Fabric network to retrieve the corresponding blockchain record. QR codes can be used to conveniently identify and access the digital drug record. The verification result is then returned to the application, allowing authorized users to view the drug's registration status, ownership, expiry information, and other relevant details.

For supply-chain transactions, authorized participants can perform operations such as drug registration and ownership transfer through the blockchain layer. The current prototype uses Hyperledger Fabric v2.5.16, with Org1 representing the Manufacturer and Org2 representing the Hospital, connected through the drugchannel channel. The drugtrace chaincode provides functions including registerDrug, readDrug, drugExists, transferDrug, verifyDrug, and getDrugHistory. When a transfer transaction is performed, the updated ownership information is recorded on the ledger, while the historical transaction record can be retrieved for audit and traceability purposes.

The database and application services maintain operational information such as user profiles, inventory records, scan logs, shipment information, notifications, and analytics data. PostgreSQL is planned as the off-chain database for information that does not need to be stored directly on the blockchain, while critical drug identity and transaction events are maintained through Hyperledger Fabric. The backend, planned using Node.js and Express, acts as the communication layer between the user interface, database, and blockchain network. Through this flow, the proposed system connects drug registration, inventory management, QR-assisted verification, ownership tracking, shipment monitoring,

blockchain transactions, alerts, and reporting into a unified pharmaceutical supply-chain workflow.

For the Level-0 Data Flow Diagram, the complete system can be represented as a single central process, Drug Inventory and Supply Chain Tracking System, interacting with external entities such as Manufacturer, Distributor/Warehouse, Hospital/Pharmacy, and Administrator. Users provide drug, inventory, verification, and transaction information to the system, while the system returns drug details, verification results, inventory status, ownership information, shipment updates, alerts, and reports.

For the Level-1 Data Flow Diagram, the central system can be divided into major processes including User and Access Management, Drug Registration and Identification, Inventory Management, Drug Verification, Shipment and Ownership Tracking, Blockchain Transaction Management, and Reporting and Notifications. These processes communicate with the PostgreSQL database and Hyperledger Fabric network according to the type of information being processed. The blockchain maintains trusted drug and transaction records, while the database stores supporting operational information required by the application.

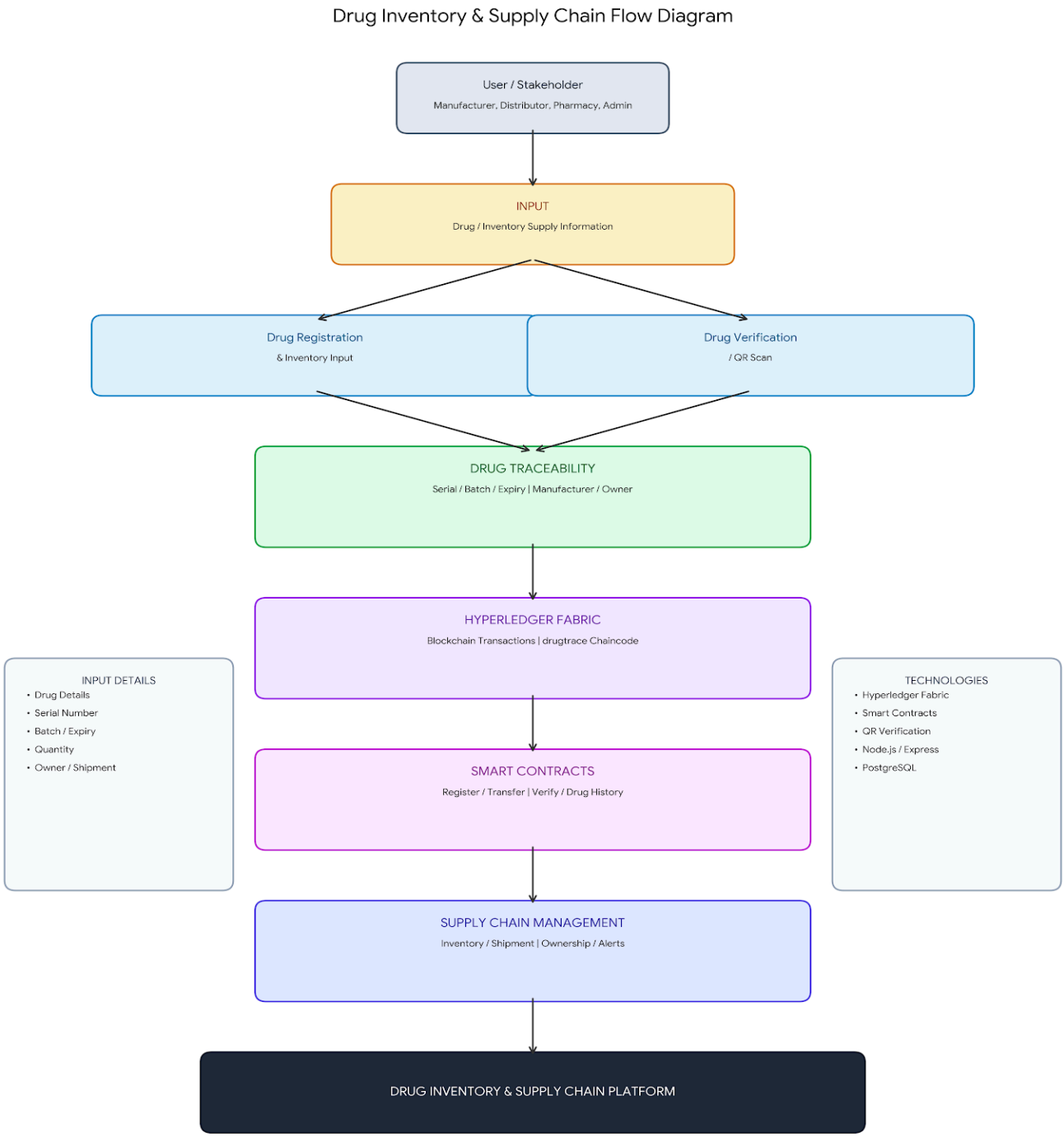


Fig 3.2: flow diagram

System Use Case Analysis

The **Drug Inventory and Supply Chain Tracking System** coordinates interactions between its primary pharmaceutical stakeholders, application services, blockchain components, and inventory-management services:

- **Manufacturer:** Uses the system to register new drug records, provide information such as serial number, batch number, drug name, manufacturing date, expiry date, quantity, and manufacturer details, and initiate the drug's supply-chain lifecycle. In the current prototype, the Manufacturer is represented by **Org1** in the Hyperledger Fabric network.
- **Distributor/Warehouse:** Uses the platform to access authorized drug information, monitor inventory, receive and transfer medicines, and update ownership or shipment-related information as the drug moves through the supply chain. These roles are part of the proposed system architecture and can be incorporated as additional organizations in future network expansion.
- **Hospital/Pharmacy:** Uses the system to verify received medicines, view drug and batch information, check expiry and ownership details, and maintain visibility of relevant inventory and supply-chain records. In the current prototype, the Hospital is represented by **Org2** in the Hyperledger Fabric network.
- **Administrator:** Manages system-level information, authorized users, organizations, inventory-related records, alerts, reports, and other application resources. The administrator can also monitor supply-chain activities and maintain appropriate access controls for different stakeholders.
- **Backend/API Layer:** Acts as the communication and business-logic layer between the user interface, database, and blockchain network. The proposed architecture uses **Node.js and Express** to process application requests, manage business logic, and communicate with the Hyperledger Fabric network through the appropriate gateway services.
- **Hyperledger Fabric Network:** Provides the permissioned blockchain infrastructure for recording and verifying critical drug transactions. The current prototype uses **Hyperledger Fabric v2.5.16**, with **Org1 (Manufacturer)** and **Org2 (Hospital)** connected through the drugchannel channel.
- **Smart Contract / Chaincode:** The drugtrace chaincode manages the core drug-tracking operations. Its current functions include **registerDrug**, **readDrug**, **drugExists**, **transferDrug**, **verifyDrug**, and **getDrugHistory**. These functions support drug registration, record retrieval, ownership transfer, verification, and historical transaction tracking.
- **PostgreSQL / Off-Chain Services:** The proposed PostgreSQL layer stores supporting operational information such as user profiles, inventory records, scan logs, shipment information, notifications, and analytics that do not need to be stored directly on the blockchain. This allows the blockchain to focus on critical drug identity and transaction records while the application manages operational data efficiently.

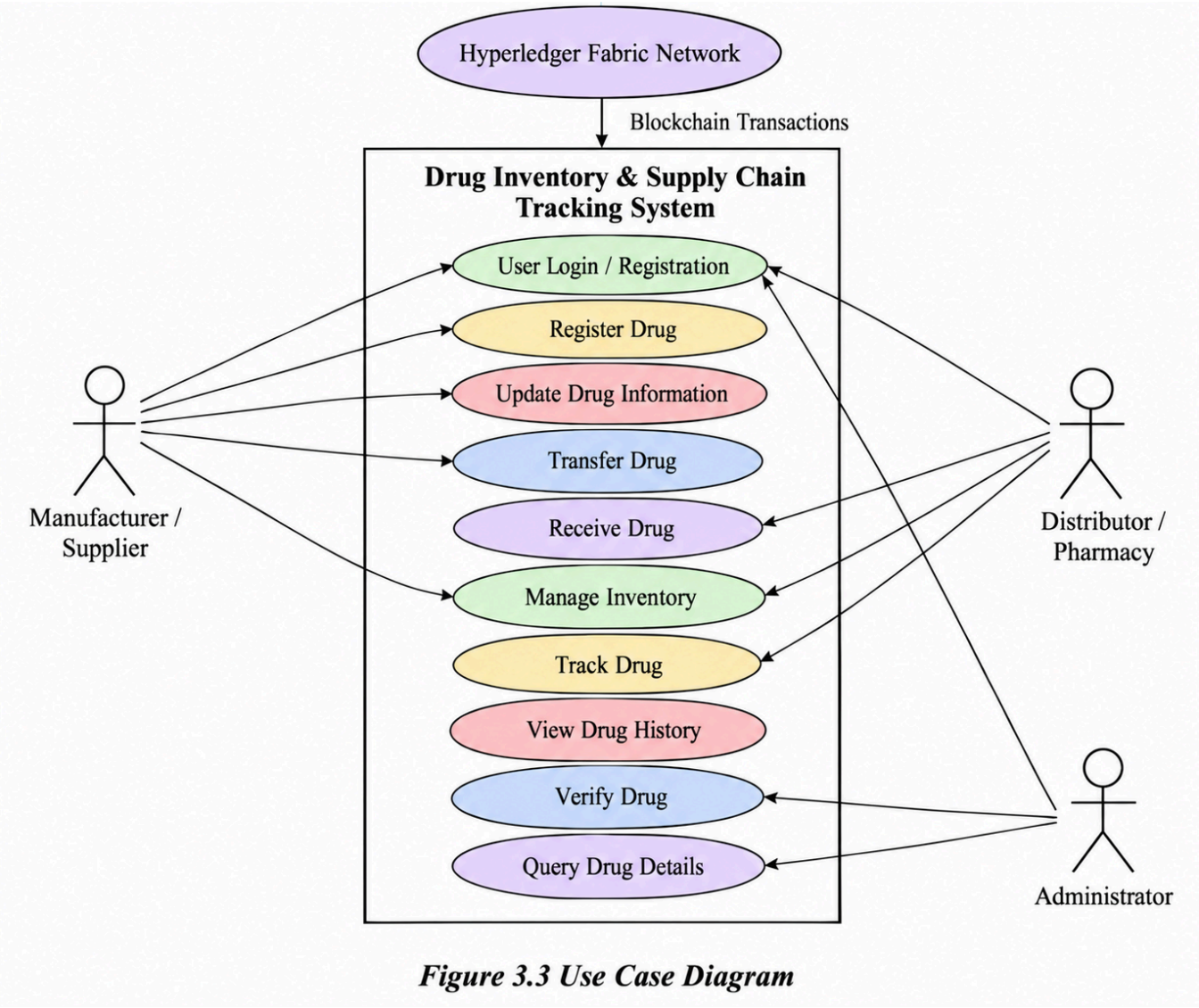


Figure 3.3 Use Case Diagram

CHAPTER 4
PROJECT DESCRIPTION

4.1METHODOLOGIES

4.1.1Drug and Stakeholder Interface Module

The Drug and Stakeholder Interface module provides the primary interaction layer between users and the Drug Inventory and Supply Chain Tracking System. It is designed to provide a simple and accessible environment through which authorized stakeholders such as manufacturers, distributors, warehouse personnel, hospitals, pharmacies, and administrators can log in, manage their profiles, register drugs, view inventory, verify medicines, track ownership, monitor shipments, and access supply-chain information. The interface acts as the central navigation point for the major functions of the application, including drug registration, inventory management, QR-assisted verification, shipment tracking, alerts, notifications, and reporting. These features allow the drug-tracking functionality to be presented as part of a broader pharmaceutical supply-chain platform rather than as an isolated blockchain service.

The module begins with a secure login and authentication interface. After successful authentication, the user is directed to the dashboard, where the major services of the system can be accessed according to the user's role. A manufacturer can register drugs and monitor their records, while hospitals and other authorized stakeholders can verify medicines, view drug information, monitor inventory, and track ownership or shipment details. The interface therefore acts as the connecting layer between users and the underlying inventory management, blockchain, verification, and supply-chain components.

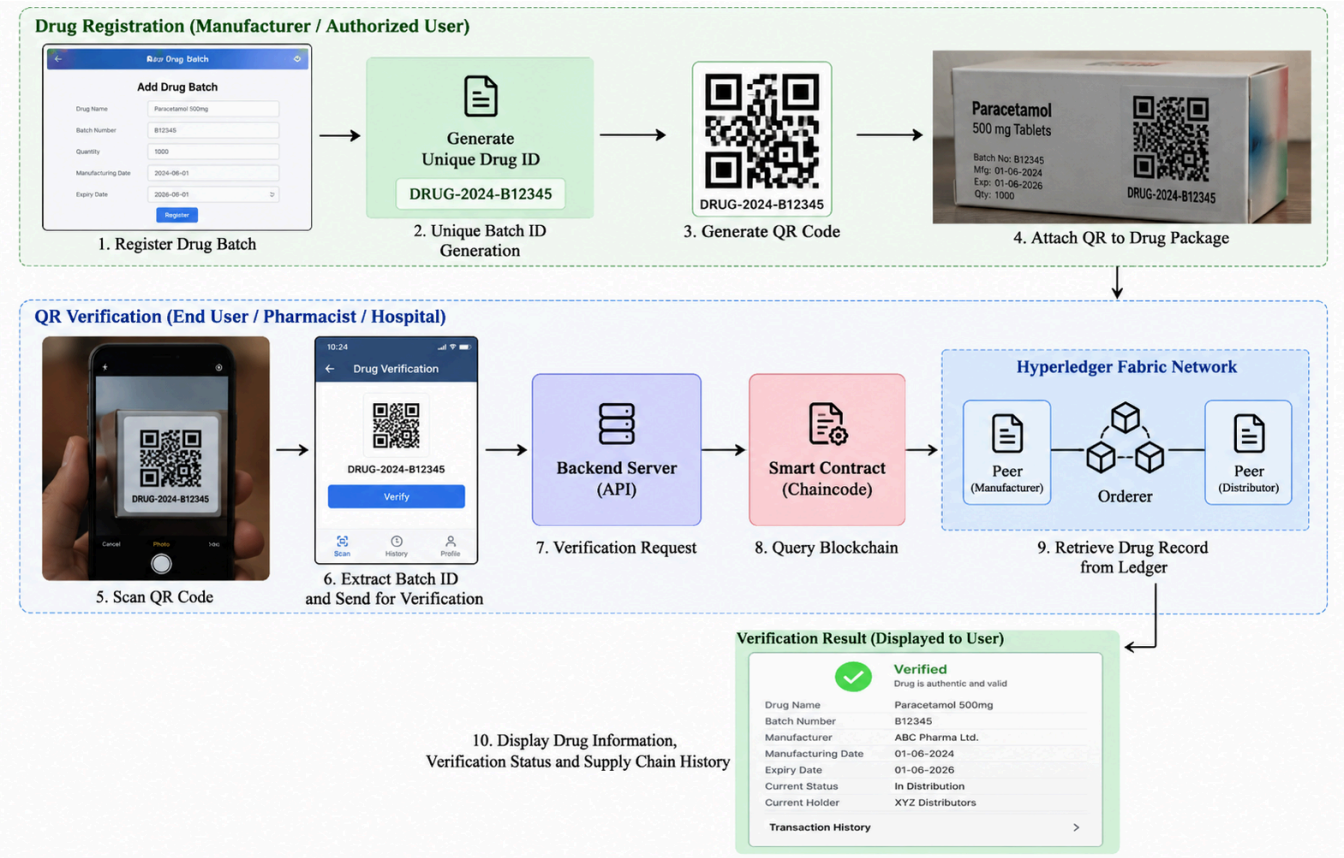


Fig 4.1: QR-Based Verification Flow

Major Functional Components

1. User Registration and Login:

The system provides a registration and login mechanism for authorized supply-chain participants. Authentication ensures that user-specific information and protected operations are associated with the correct account.

2. Role and Profile Management:

Users are associated with their respective roles, such as manufacturer, distributor, warehouse, hospital, pharmacy, or administrator. Profile information provides the basis for controlling access to supply-chain operations.

3. Dashboard:

The dashboard provides a centralized view of the major system functionalities. Users can access drug registration, inventory, verification, shipment tracking, alerts, reports, and other available services according to their role.

4. Drug Registration:

Authorized manufacturers can enter drug details such as serial number, batch number, drug name, manufacturer, manufacturing date, expiry date, and quantity for registration in the system.

5. Drug Verification:

Users can verify a drug using its unique serial number and access its corresponding digital record. QR-based identification can be used as a convenient method for accessing the drug record.

6. Inventory Management:

The interface allows authorized users to view and manage available drug stock, batch information, expiry information, ownership, and drug status.

7. Shipment and Ownership Tracking:

Users can monitor the movement and ownership of drugs between supply-chain participants. Ownership transfers are recorded through blockchain transactions.

8. Alerts and Notifications:

The system can provide notifications related to low stock, approaching expiry dates, shipment activities, and other important supply-chain events.

9. Reports and History:

Authorized users can access drug transaction history, ownership information, verification records, inventory information, and other reports required for monitoring and auditing.

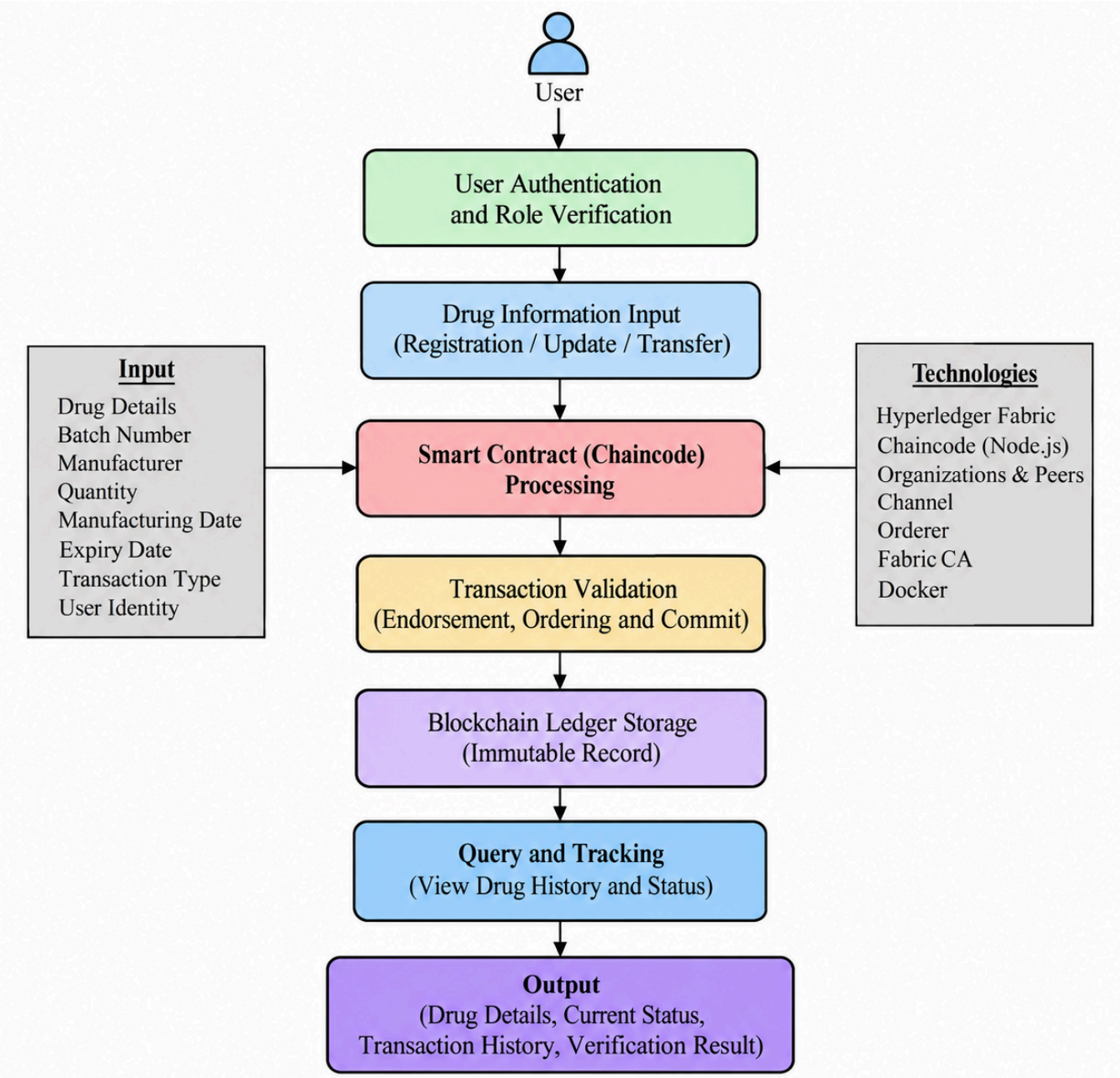


Figure 4.1: Methodology Flowchart

4.2 MODULE DESCRIPTION

4.2.1 Drug and Stakeholder Interface Module

The Drug and Stakeholder Interface module provides the primary interaction layer between users and the Drug Inventory and Supply Chain Tracking System. It is designed to provide a simple and accessible environment through which authorized stakeholders such as manufacturers, distributors, warehouse personnel, hospitals, pharmacies, and administrators can log in, manage their profiles, register drugs,

view inventory, verify medicines, track ownership, and access supply-chain information. The dashboard acts as the central navigation point for the major functions of the application. The module also incorporates inventory information, QR-assisted verification, shipment tracking, alerts, notifications, and transaction history so that drug-tracking functionality is presented as part of a broader pharmaceutical supply-chain platform rather than as an isolated blockchain service.

The module begins with a secure login and authentication interface. After successful authentication, the user is directed to the dashboard, where the major services of the system can be accessed according to the user's role. The manufacturer can register drugs and manage drug records, while hospitals and other authorized stakeholders can verify medicines, view drug information, monitor inventory, and track ownership or shipment details. The interface therefore acts as the connecting layer between the user-facing services and the underlying drug inventory, blockchain, verification, and supply-chain components.

4.3 Major Functional Components

1. User Registration and Login:

The system provides a registration and login mechanism for authorized supply-chain participants. Authentication ensures that user-specific information, inventory activities, verification records, and other protected resources can be associated with the correct account.

2. Stakeholder Profile Management:

After authentication, the user can maintain profile-related information required by the application. The profile identifies the stakeholder and provides the basis for associating supply-chain activities and transactions with the appropriate user or organization.

3. Dashboard:

The dashboard provides a centralized view of the major system functionalities. From the dashboard, users can access drug registration, inventory management, drug verification, shipment tracking, notifications, reports, and other available services according to their role.

4. Drug Registration:

Authorized users, particularly manufacturers, can register a drug by providing information such as serial number, batch number, drug name, manufacturer, manufacturing date, expiry date, and quantity.

5. Drug Verification:

The interface provides a mechanism through which users can verify a drug using its unique serial number. The corresponding digital record can be retrieved from the blockchain to check important drug information and status.

6. QR-Assisted Verification:

A QR-based identification mechanism can be used to provide convenient access to the digital drug record. The QR code acts as an identifier or lookup mechanism and is not considered, by itself, proof of physical authenticity.

7. Inventory and Supply-Chain Support:

The interface allows authorized stakeholders to view drug quantities, batch information, expiry details, current ownership, and drug status. It also provides access to relevant supply-chain information.

8. Shipment and Ownership Tracking:

Users can monitor the movement and ownership of medicines between supply-chain participants. Ownership changes can be recorded through blockchain transactions.

9. Alerts, Notifications and History:

The interface can provide stock and expiry alerts, shipment-related notifications, and access to previous drug transactions and verification records for monitoring and auditing purposes.

Drug and Inventory Management Module

The **Drug and Inventory Management module** forms the core inventory and pharmaceutical information layer of the blockchain-based drug supply chain system. Its primary purpose is to maintain and present structured information about medicines throughout their movement across the supply chain. The module manages drug details such as drug name, batch number, manufacturer, quantity, manufacturing date, expiry date, and current status.

The module is integrated with the **Hyperledger Fabric blockchain network**, where important drug transactions and ownership or custody changes can be recorded in a tamper-resistant and traceable manner. Each drug batch can be uniquely identified and tracked as it moves from the manufacturer through distributors and other authorized participants in the supply chain.

When a drug is registered in the system, its relevant information is recorded and associated with a unique batch or drug identifier. Subsequent activities such as transfer, receipt, inventory updates, and

status changes can be recorded as blockchain transactions. This provides authorized participants with a reliable history of the drug's movement and helps improve transparency and traceability within the pharmaceutical supply chain.

4.2.2 Major Functional Components

1. Drug Information Management:

The module maintains information about registered drugs, including drug name, drug ID, batch number, manufacturer, and other relevant details.

2. Batch Management:

Each drug batch is identified using a unique batch identifier. Batch-level information can be used to distinguish different manufacturing lots and track their movement through the supply chain.

3. Manufacturer Information:

The system maintains information about the organization or participant responsible for registering or manufacturing the drug.

4. Quantity and Inventory Management:

The module maintains the available quantity of drugs and supports inventory-related updates when drugs are received, transferred, or distributed.

5. Manufacturing and Expiry Information:

Important dates such as manufacturing date and expiry date are maintained for each drug batch. This information helps participants verify the validity of medicines.

6. Drug Status Management:

The current status of a drug or batch can be maintained to indicate its state within the supply chain, such as registered, available, transferred, received, or expired.

7. Supply Chain Tracking:

The movement of a drug batch between authorized participants can be recorded as blockchain transactions. This enables participants to trace the history of a drug throughout the supply chain.

8. Transaction History:

The system maintains the transaction history associated with a drug or batch. Authorized users can use this information to verify previous transfers and activities.

9. Drug Verification:

The blockchain-based records allow authorized participants to verify whether the drug information and its transaction history are consistent with the records stored on the network.

10. Search and Filtering:

Users can search for drugs or batches and apply available filters to locate relevant inventory and supply-chain records efficiently.

11. Inventory View:

The module provides a consolidated view of available drugs and their relevant batch information, allowing supply-chain participants to monitor inventory.

12. Detailed Drug View:

The drug-detail interface brings together the available drug, batch, inventory, and tracking information into a single view for easier verification and management.

Blockchain and Smart Contract Module

The Blockchain and Smart Contract module forms the primary computational and

transaction-processing component of the drug inventory and supply chain tracking system. It is responsible for securely recording drug-related transactions and maintaining a tamper-resistant history of drug movement across the supply chain. The module uses Hyperledger Fabric to provide a permissioned blockchain environment in which authorized participants can register, update, transfer, and verify drug information.

The module uses smart contracts (chaincode) to define the operations that can be performed on drug records. When a participant performs an operation such as registering a drug, transferring a drug batch, updating its status, or querying its history, the corresponding transaction is processed through the blockchain network. The resulting ledger records provide traceability and transparency throughout the drug supply chain without requiring a single centralized authority to maintain the complete transaction history.

The implementation uses the Hyperledger Fabric network with participating organizations, peers, an ordering service, channels, and deployed chaincode. Drug information is represented as blockchain assets, while smart-contract functions control how these assets are created, accessed, updated, and queried.

4.2.3 Major Functional Components

1. Drug Registration

The module allows an authorized participant to register a new drug or drug batch on the blockchain. The drug record can contain information such as drug ID, drug name, batch number, manufacturer, quantity, manufacturing date, and expiry date.

2. Unique Drug and Batch Identification

Each drug or batch is associated with a unique identifier. This identifier is used to distinguish individual records and to perform subsequent blockchain transactions.

3. Smart Contract Processing

Smart contracts define the business logic for managing drug records. Operations such as registration, transfer, updating, and querying are executed according to the rules implemented in the chaincode.

4. Drug Transfer:

The module records the transfer of a drug or batch from one authorized participant to another.

Each successful transfer becomes part of the drug's transaction history.

5. Ownership / Custody Management:

The current participant responsible for a drug or batch can be maintained in the blockchain record. This helps identify the current location or custody of the drug within the supply chain.

6. Inventory Updates:

Drug quantities and relevant inventory information can be updated through authorized blockchain transactions when drugs are received, transferred, or otherwise processed.

7. Drug Status Management:

The status of a drug or batch can be maintained throughout its lifecycle. This can be used to represent conditions such as registered, available, transferred, received, or expired.

8. Transaction History:

The blockchain maintains the history of transactions associated with a drug. Authorized participants can query this history to determine how the drug has moved through the supply chain.

9. Drug Verification:

The module enables authorized participants to query blockchain records and verify the stored information about a drug, including its identity, current status, and transaction history.

10. Query and Retrieval:

Smart-contract query functions allow users or authorized applications to retrieve individual drug records, retrieve multiple assets, and obtain relevant supply-chain information from the ledger.

11. Blockchain Consensus and Ordering:

Transactions submitted to the network are processed through Hyperledger Fabric's transaction flow, including endorsement, ordering, and validation before being committed to the ledger.

12. Immutable Ledger Storage:

Once valid transactions are committed, they form part of the blockchain ledger. This provides a reliable historical record that can be used for auditing and traceability.

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

DrugTrace is a blockchain-based Drug Inventory and Supply Chain Tracking System designed to provide secure, transparent and traceable management of pharmaceutical drugs throughout the supply chain. The system uses Hyperledger Fabric to maintain tamper-resistant drug records and track important information such as serial number, batch number, drug name, manufacturer, manufacturing date, expiry date, quantity and current ownership. A dedicated drugchannel and drugtrace chaincode manage drug registration, verification, ownership transfer and transaction history across the blockchain network. The system provides REST-based integration through a Node.js and Express backend, which acts as the gateway between the application interfaces and Hyperledger Fabric. The web application provides modules for drug registration, inventory management, drug verification, ownership transfer and blockchain transaction history. A QR-based identification mechanism generates a unique QR code containing the drug serial number, which can be attached to a drug package for quick verification. The Android mobile application scans the QR code and communicates with the backend to retrieve the corresponding blockchain-backed drug record, allowing users to verify drug details, current ownership and status. The system follows a layered architecture consisting of the React web application, Android mobile application, Node.js REST backend and Hyperledger Fabric network, ensuring that blockchain credentials and network configuration remain protected from client applications. Overall, DrugTrace provides a secure and transparent approach to pharmaceutical inventory and supply-chain tracking by combining blockchain-based traceability, ownership management, QR-based verification and mobile accessibility.

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